Autonomous Multi-Agent AI Data Scientist

Major Project (B.Tech 4th Year) | Ekta (B23CS1018), Kashish Joshi (B23CS1025), Kurra Hema (B23CS1031)

Analytical Goal: predict whether tumor is malignant based on measurements

Audit Date: 11 November 2026 | Dataset: 569 Rows x 31 Columns | Evidence Confidence: 99%

1. Executive Summary

An autonomous end-to-end multi-agent evaluation was executed on the dataset using a LangGraph StateGraph pipeline. The best performing architecture was LogisticRegression achieving a holdout test score of 0.9825 (5-fold CV: 0.978 ± 0.0098) with an overfitting generalization gap of 0.7%. The Critic Agent completed a 4-pillar verification audit and certified the findings at a 99% Evidence Confidence Score.

2. Machine Learning Benchmark Leaderboard (5-Fold Stratified CV)

Model Architecture	5-Fold CV Score	Holdout Acc/R2	Train Score	F1 / RMSE	Overfit Gap
RandomForest	0.9626	0.9474	0.9956	0.9474	4.8%
GradientBoosting	0.9516	0.9474	1.0	0.9471	5.3%
ExtraTrees	0.956	0.9474	0.989	0.9471	4.2%
LogisticRegression	0.978	0.9825	0.989	0.9825	0.7%

3. Explainable AI: Top Predictive Determinants

#1 worst area (Importance: 13.7% | Direction: Protective / Negative (-))
#2 worst concave points (Importance: 12.8% | Direction: Protective / Negative (-))
#3 worst radius (Importance: 10.0% | Direction: Protective / Negative (-))
#4 mean concave points (Importance: 9.2% | Direction: Protective / Negative (-))
#5 worst perimeter (Importance: 7.5% | Direction: Protective / Negative (-))

4. Prioritized Strategic Action Plan

[P0] Prioritize Biometric Triage on Top Morphological Markers

Statistical models confirm 'worst area' exhibits the highest discriminatory power. Prioritize histological review of these metrics in automated pathologist triage workflows.

Project

[P1] Dual-Practitioner Verification for Borderline Probabilities

Implement an automated protocol routing predictions with model classification probabilities between 0.40 and 0.60 for secondary biopsy verification.

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5. Critic Agent 4-Pillar Empirical Validation Audit (Confidence: 99%)

- Pillar 1 Verified: Statistical power confirmed with 569 records (Train=455, Holdout=114).

- Pillar

6. Publication Visualizations & Empirical Artifacts







